Chapter test

22 Respiration

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

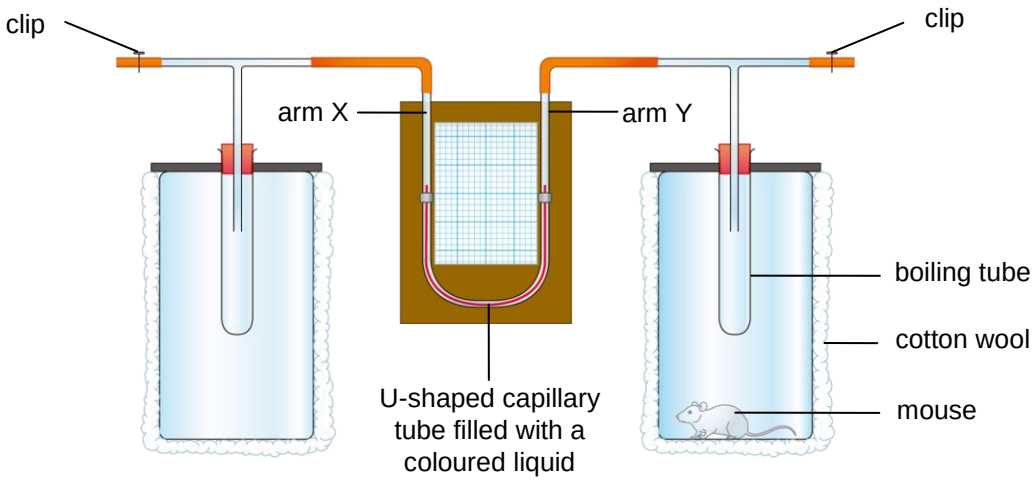
Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following cells probably have the lowest rate of respiration?

   A. liver cells
   B. muscle cells
   C. epithelial cells of air sacs of the lungs
   D. root hair cells

Directions: Questions 2 and 3 refer to the diagram below, which shows a set-up used to study respiration of a mouse.



2. What is the purpose of this study?

   A. to show that oxygen is consumed in respiration
   B. to show that carbon dioxide is produced in respiration
   C. to show that heat is released in respiration
   D. to show that water vapour is released in respiration

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3. What would be the change in the liquid level at arms X and Y of the U-shaped capillary tube after 15 minutes?

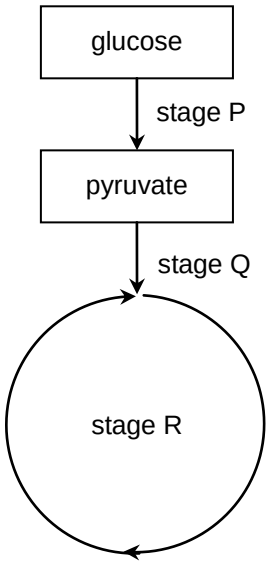
	At arm X	***At arm Y***
A.	no change	rise
B.	no change	drop
C.	drop	rise
D.	rise	drop

4. Which of the following is / are necessary for glycolysis to occur?

(1) ATP
(2) oxygen
(3) NADH

A. (1) only
B. (2) only
C. (1) and (2)
D. (1) and (3)

5. The diagram below shows some stages of aerobic respiration.



At which stage(s) is carbon dioxide released?

A. R only
B. P and R only
C. P and Q only
D. Q and R only

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6. Which of the following statements about pyruvate is / are correct?

(1) It contains less energy than glucose per molecule.
(2) It is produced in the mitochondria.
(3) It is converted into lactic acid when oxygen is absent.

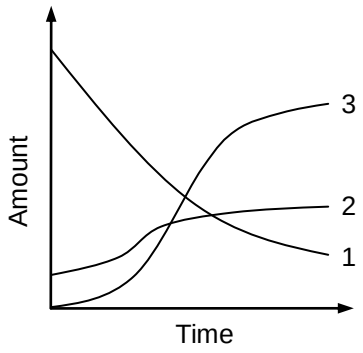
A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

7. Which of the following is / are produced by both aerobic respiration and anaerobic respiration in humans?

(1) ATP
(2) water
(3) carbon dioxide

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

★ 8. In a set-up, some living yeast cells were added to glucose solution that had been boiled and cooled. The graph below shows the changes in the amounts of three substances in the set-up.



What do curves 1, 2 and 3 represent, respectively?

	1	***2***	***3***
A.	yeast	glucose	ethanol
B.	glucose	yeast	ethanol
C.	ethanol	yeast	glucose
D.	glucose	ethanol	yeast ☐

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9. Which of the following comparisons between alcoholic fermentation and lactic acid fermentation is correct?

	Alcoholic fermentation	*Lactic acid fermentation*
A.	uses alcohol as the substrate	uses lactic acid as the substrate
B.	more energy is released	less energy is released
C.	produces carbon dioxide	does not produced carbon dioxide
D.	occurs in yeast only	occurs in animal only

10. Which of the following comparisons between aerobic respiration and photosynthesis is correct?

	Aerobic respiration	*Photosynthesis*
A.	anabolic process	catabolic process
B.	all stages occur in mitochondria	all stages occur in chloroplasts
C.	electron transport chain involved	electron transport chain not involved
D.	forms water in the final stage	uses water in the first stage

Question	1	2	3	4	5	6	7	8	9	10
Related section	22.1	22.3	22.3	22.3	22.3	22.5	22.5	22.4	22.4	22.6

B. Problem solving (22 marks)

11. The photomicrograph below shows a section through a mitochondrion.



(a) Identify the structure labelled X and explain how it is adapted for the organelle to function efficiently. (3 marks)

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(b) Mitochondria are variable in shape and size. Their length varies from 1.5 to 10.0 μm. However, their diameter does not exceed 1.0 μm. Suggest the significance of mitochondria having a relatively small diameter. (1 mark)

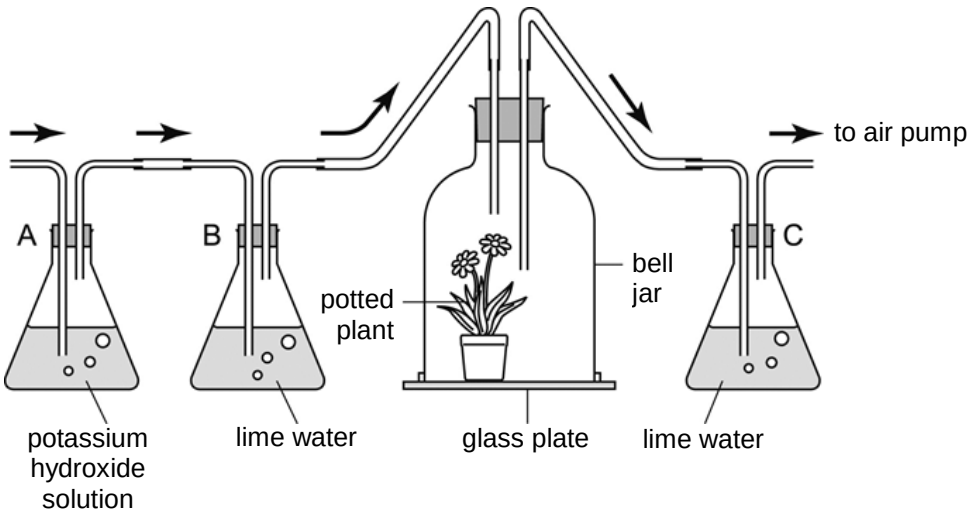
(c) State ***two*** chemical substances that

(i) enter mitochondria from the cytoplasm during respiration. (2 marks)

(ii) leave mitochondria and enter the cytoplasm during respiration. (2 marks)

(Total: 8 marks)

★ 12. The diagram below shows an experimental set-up used to study respiration of a potted plant.



(a) State the function of

(i) the potassium hydroxide solution in flask A. (1 mark)

(ii) the lime water in flask B. (1 mark)

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(b) The apparatus has not been set up properly. State, with reasons, ***two*** corrections that should be made to correct the set-up. (4 marks)

(c) What results would be observed after the correction? What do the results show? (2 marks)

(Total: 8 marks)

★ 13. Below are the steps in bread-making.

(1)	Mix flour, sugar, water and yeast to form a dough.
(2)	Leave the dough undisturbed at 30 °C for 1 hour.
(3)	Bake the dough in an oven at 230 °C for 30 minutes.

(a) Name the process carried out by yeast that can make the dough rise in step (2). Explain why the process can make the dough rise. (3 marks)

(b) The dough will not rise any further after it has been put in the oven for 10 minutes. Suggest a reason to explain why. (1 mark)

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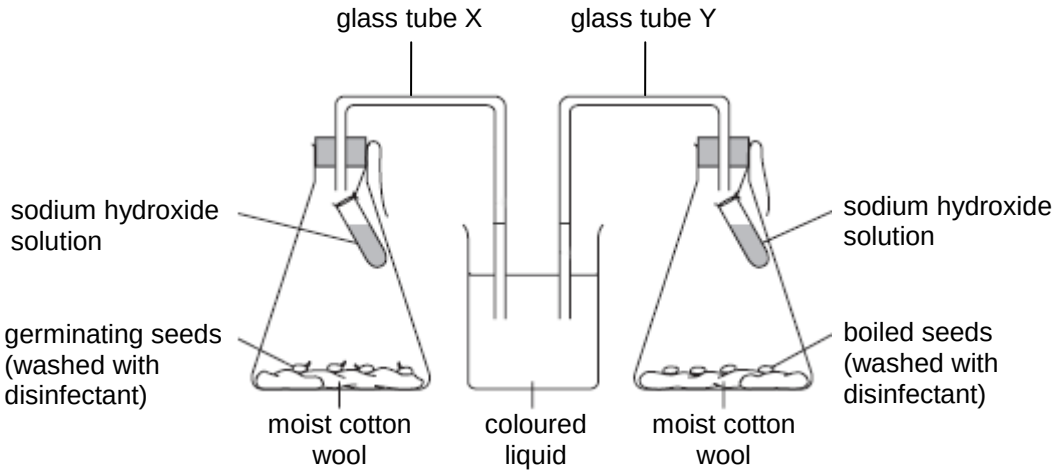
(c) (i) Name another product produced by the yeast during bread-making. (1 mark)

(ii) What happens to this product when the dough is baked? (1 mark)

(Total: 6 marks)

C. Exam challenge (8 marks)

★★ 14. The diagram below shows an experimental set-up used to measure the rate of respiration of some geminating seeds.



(a) Two hours after the start of the experiment, the liquid level in glass tube X rose for 1 cm and the liquid level in glass tube Y dropped for 1 cm.

(i) Explain the changes in the liquid levels. (4 marks)

(ii) Given that the cross-sectional area of the glass tubes is 0.5 cm^2, calculate the rate of respiration of the germinating seeds. (1 mark)

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(b) The rise in the liquid level in glass tube X gradually slowed down. Explain why. (2 marks)

(c) Explain why the germinating seeds and the boiled seeds should be washed with disinfectant before the experiment. (1 mark)

(Total: 8 marks)

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Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	C	2.	C	3.	D	4.	A	5.	D
6.	C	7.	A	8.	B	9.	C	10.	D

(1 mark each)

B. Problem solving (22 marks)

11. (a) Inner membrane / Cristae (1)
It is highly folded to provide a large surface area (1)
for the attachment of enzymes / carriers involved in aerobic respiration. (1)
(b) To shorten the diffusion distance for rapid exchange of materials (1)
(c) (i) Any ***two*** of the following: (2)
oxygen / pyruvate / NADH (reduced NAD) / ADP / phosphate
(ii) Any ***two*** of the following: (2)
carbon dioxide / NAD / ATP / water

(Total: 8 marks)

12. (a) (i) To remove carbon dioxide from the air (1)
(ii) To show that the air entering the bell jar is free of carbon dioxide (1)
(b) The bell jar should be covered with a piece of black cloth in order to exclude light. (1)
Otherwise, the potted plant will undergo photosynthesis and absorb carbon dioxide. (1)
The pot should be wrapped in a polythene bag (1)
to prevent the respiration of soil microorganisms from affecting the result. (1)
(c) The lime water in flask C would turn milky. (1)
This shows that that potted plant releases carbon dioxide. (1)

(Total: 8 marks)

13. (a) Fermentation / Alcoholic fermentation / Anaerobic respiration (1)
The process produces carbon dioxide (1)
which is trapped inside the dough (1)
making it rises.
(b) The yeast are killed / enzymes are denatured under high temperature. (1)
Thus no more carbon dioxide is produced.
(c) (i) Ethanol / Alcohol (1)
(ii) It is evaporated by the heat of the oven. (1)

(Total: 6 marks)

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C. Exam challenge (8 marks)

14. (a) (i) Germinating seeds consumed oxygen during respiration and released carbon dioxide. (1)

The carbon dioxide produced was absorbed by the sodium hydroxide solution. (1)

The air pressure in the flask containing the germinating seeds became lower than the atmospheric pressure. (1)

As a result, the coloured liquid in glass tube X was drawn upwards.

The liquid level in glass tube Y dropped because of the weight of the liquid. (1)

(ii) $(1 + 1) \times 0.5 / 2 = 0.5\ cm^3$/hour (1)

(b) As oxygen in the flask became used up, germinating seeds switched to anaerobic respiration. (1)

Less and less (No more) carbon dioxide was produced. (1)

(c) This is to kill any microorganisms on the surface of the seeds. Otherwise, microorganisms may carry out respiration and affect the results. (1)

(Total: 8 marks)

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